Project Title: Root Cause Analysis Model for Semiconductor Mass Production Failures
Course: APAN PS5900 – Capstone: Solving Real World Problems with Analytics
Client: Renesas Electronics
Team Members: Ashish Mathew, Bowen Gu, Huan Hsu, Marielle Alexander, Miguel Morales, Oluwafisayo Ojo, Tingting Fang, Xiaohua Liu, Yaritza Soto
Title: Project Process Flow Breakdown

This document outlines the full analytical and modeling workflow (summarized visually in Figure 1.) used to build a multivariate Root Cause Analysis (RCA) pipeline for Renesas semiconductor mass-production data. Each step is described clearly to ensure that the analysis can be fully replicated by the sponsor or by future teams. The process begins with a single comprehensive Renesas dataset and concludes with an interpretable decision-tree RCA model that generates threshold-based rules for identifying and understanding failure mechanisms.

1. Start of Analysis: Raw Dataset

The project begins with loading the complete Renesas dataset, which contains WT1 and WAT electrical tests, die-level metadata, wafer and lot identifiers, probe card IDs, and final pass/fail labels. No external datasets are used. This ensures full reproducibility from a single source of truth.

2. Data Pre-Processing

Pre-processing prepares the dataset for modeling and ensures that downstream results are reliable.

Steps performed:

- **Filter to Category C failures** since these were identified by Renesas as the most operationally important failure class.

- **Remove features with more than 10% missing values**, as these tests are unreliable for machine learning and produce instability across folds.

- **Handle structured missingness** by applying median imputation for WT1/WAT tests and retaining missingness indicators where they carry operational meaning.

- **Apply feature standardization** (scaling) for all models except the decision tree, which requires raw values for interpretability.

- **Use class weighting (1:10)** to adjust for the strong imbalance in the dataset (~15% failures).

These steps ensure that the data is clean, consistent, and ready for feature-selection modeling.

3. Feature Selection & Modeling (Lasso, Random Forest, XGBoost)

The next stage involves training three complementary feature-selection models using the same preprocessed dataset.

Modeling setup:

- All models are trained using **5-Fold Cross-Validation**.
- **RandomizedSearchCV** is used for hyperparameter tuning to avoid overfitting and to handle the high dimensionality of tests.
- Each model returns a ranked list of the most important test parameters.

Models used:

- **Lasso (L1)**: captures linear associations and performs embedded feature selection.
- **Random Forest:** identifies nonlinear interactions and computes impurity-based importance.
- **XGBoost:** captures complex boosted patterns and ranks features by split gain.

Using three different methods ensures robustness and reduces dependence on any single modeling approach.

4. Extract Feature Importance from Each Model

After training, each model produces its own importance scores. These outputs are stored and standardized before aggregation.

5. Aggregation of Feature Importances

To create a stable and reliable ranking, importances from all three models are **normalized**, then **averaged** across models.

This step:

- Reduces model-specific noise
- Highlights tests consistently influential across all methods
- Produces a unified, reproducible ranking list

The aggregated ranking identified **WAT_30_polar** as the strongest overall predictor.

6. Select Top 50 Features

The aggregated importance scores are sorted, and the **top 50 features** are selected for the RCA model.

This limits noise, improves interpretability, and stabilizes thresholds in the decision tree.

7. Multivariate RCA: Decision Tree Model

A decision-tree classifier is trained using the selected top 50 features.

Configuration:

- **No feature scaling**, ensuring thresholds are directly interpretable.
- **Class weighting (1:10)** is maintained to offset class imbalance.
- **RandomizedSearchCV** tunes depth, splitting criteria, and node sizes.

Why a decision tree?
It provides:

- Clear, rule-based splits
- Human-interpretable thresholds

- Visual nodes showing distinct failure vs pass regions (blue = pass, orange = fail, darker = more failure concentration)

This step produces the primary RCA output valued by Renesas.

8. Threshold-Based Rules & RCA Insights

The final decision tree is analyzed to extract:

- Test thresholds separating passing and failing dies
- Combinations of WT1/WAT tests that strongly predict failures
- Deep vs shallow node behavior to identify rare but impactful mechanisms
- High-risk zones where multiple failure indicators converge

These insights form the foundation for engineering actions such as adjusting test limits, monitoring probe card degradation, or inspecting tool-specific drift.

9. End of Process: Replicable RCA Pipeline

The workflow concludes with a reproducible, end-to-end modeling pipeline that can be operationalized by Renesas. The process is fully traceable, from dataset loading to RCA threshold extraction, and is structured so each step can be replicated, audited, or modified for new product lines.

Summary

This project flow produces a complete, interpretable RCA capability by systematically transforming raw Renesas data into meaningful engineering insights. Each stage from preprocessing to ensemble feature ranking to decision-tree threshold extraction is designed to be transparent, reproducible, and directly applicable to manufacturing root-cause investigations.

Appendix: Figure Captions
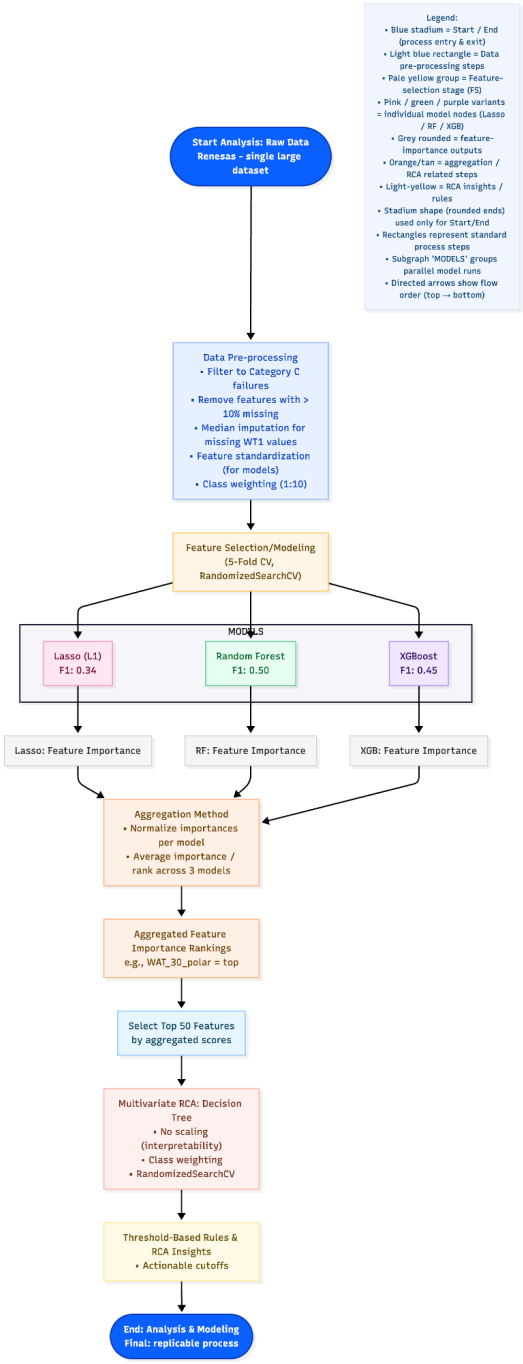
Figure 1. Visual representation of our process flowchart.